Vector India | ***Trainee – Advanced Embedded Systems***

August 2023 – February 2024

- Developed bare-metal drivers and firmware in C for ARM-based microcontrollers (LPC2148/STM32) handling UART, SPI, and I2C peripherals.
- Configured FreeRTOS tasks, queues, semaphores, and mutexes for reliable real-time task scheduling.

KEY PROJECTS

Redundant Controller Failover System (ESP32 / FreeRTOS)

- Designed a dual-controller hardware failover architecture using FreeRTOS on ESP32.
- Configured a custom pulse-based heartbeat (100ms interval, 500ms debounce threshold) over Bluetooth and established inter-module synchronization via UART to handle seamless failover without system downtime.

Smart Gate Keeping System (Automated Surveillance)

- Built an automated vehicle monitoring pipeline integrating Embedded Linux modules and computer vision models.
- Optimized hardware resource usage and peripheral communication for real-time edge processing.

Face Mask Detection (ESP32 Edge AI)

- An ESP32-based Edge AI system that detects with mask, without mask, and incorrectly worn mask from images captured through a web interface.
- The system uses a lightweight INT8 TensorFlow Lite Micro model for on-device inference, with Wi-Fi and an HTTP/HTTPS server for real-time predictions.

EDUCATION

B.Tech in Electronics and Communication Engineering

Indian Institute of Information Technology Design and Manufacturing (IIITDM), Kurnool (2019 – 2023)